

25/09/25

Dimensionality Reduction

Principal Component Analysis (PCA)

Feature selection

$f_1, f_2, \dots, f_d$

drop f_1 , keep $f_2 \rightarrow f_d - f_1$

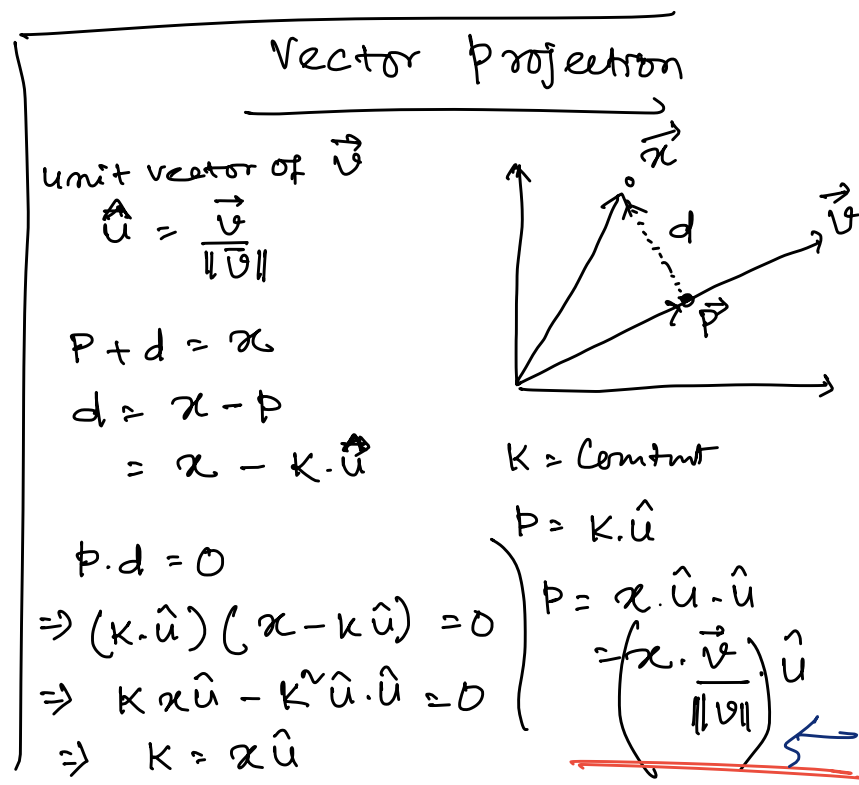
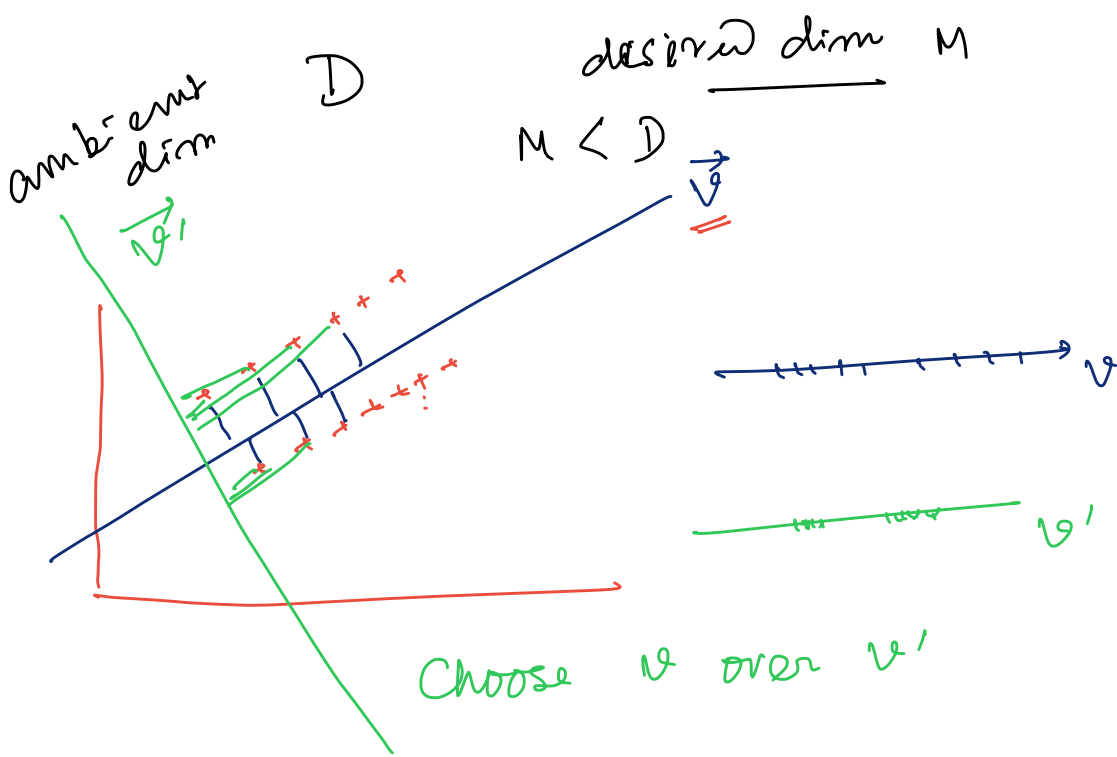
drop f_2 , keep $f_1, f_3, \dots, f_d - f_2$

original dimension $\rightarrow$ Ambient dimensionality

$\rightarrow$ Intrinsic dimensionality
the possible lower dimensions that we can project

Normalization and dimensionality reduction follow one after another

- Better visualization of data points
- Scrutinize data points manually more effectively
- Speed up the supervised learning methods
- Get rid of noise



$\{x_n\}_{n=1 \dots N}$ $\Rightarrow$ want to project them to $M < D$
 we choose such an axis that will maximize variance of the projected data points.

projection of a data x_i to v is $\text{proj}(x_i) = v^T x_i \cdot \hat{v}$
 mean project is : $v^T \bar{x} \cdot \hat{v}$

Variance of projects.

$$\begin{aligned} & \frac{1}{N} \sum_{i=1}^N (v^T x_i - v^T \bar{x})^2 \\ &= \frac{1}{N} \sum_{i=1}^N [v^T (x_i - \bar{x})]^2 \\ &= \frac{1}{N} \sum_{i=1}^N [v^T (x_i - \bar{x}) (x_i - \bar{x})^T v] \\ &= \frac{1}{N} v^T \left[\sum_{i=1}^N (x_i - \bar{x}) (x_i - \bar{x})^T \right] v \\ &= \frac{1}{N} v^T S v \end{aligned}$$

$S =$ the covariance matrix

$\max \underbrace{(v^T S v)}_{\text{best direction} = \text{best unit vector}} \text{ s.t. } \underline{\underline{v^T v = 1}}$

$$\begin{aligned} \mathcal{L} &= v^T S v - \lambda (v^T v - 1) \\ &= v^T S v + \lambda (1 - v^T v) \end{aligned}$$

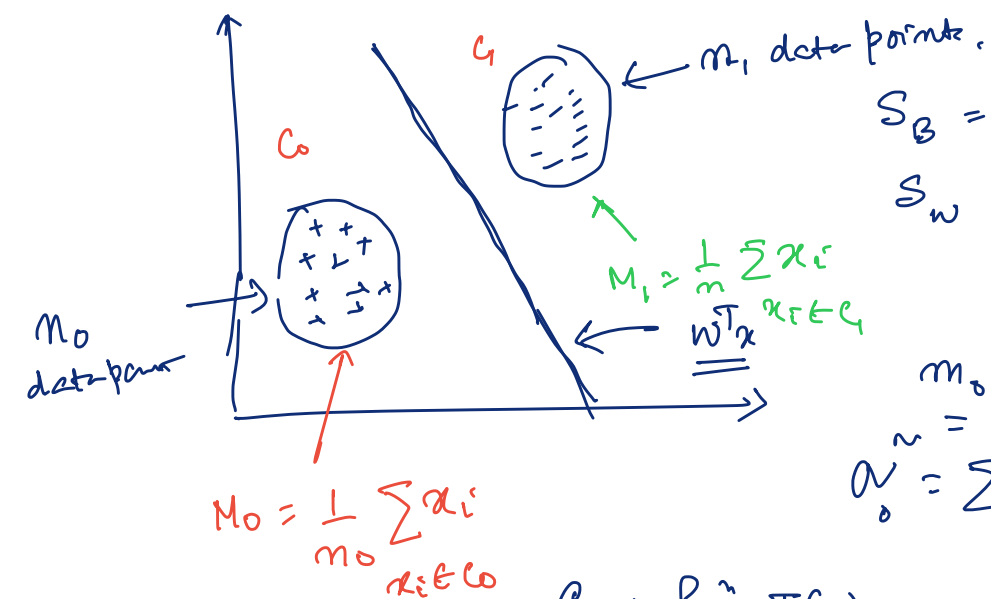
$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial v} &= 2 v S - 2 \lambda v = 0 \\ S v &= \lambda v \end{aligned}$$

Inner product of v^T to LHS and RHS

$$\underbrace{(v^T S v)}_{\text{best direction} = \text{best unit vector}} + \lambda \underline{\underline{v^T v}} = \underline{\underline{\lambda}}$$

The principal component of the data points is the largest ~~length~~ eigenvector of S

Linear Discriminant Analysis (LDA)



S_b = Between class separability

S_w = in class separability

$$m_0 = w^T M_0 \quad m_1 = w^T M_1$$

$$\sigma_0^2 = \sum (w^T x_i - m_0)^2 \quad \sigma_1^2 = \sum (w^T x_i - m_1)^2$$

$$\text{Cost fn } J(w) = \frac{(m_0 - m_1)^2}{\sigma_0^2 + \sigma_1^2} = \frac{w^T S_b w}{w^T S_w w}$$

$$(m_0 - m_1)^2 = (w^T M_0 - w^T M_1)^2$$

$$= w^T [(M_0 - M_1)(M_0 - M_1)^T] w$$

$$= w^T S_b w$$

$$\sigma_0^2 = \sum (w^T x_i - m_0)^2 = \sum (w^T x_i - w^T M_0)^2$$

$$= w^T \left[\sum_{i=1}^{n_0} (x_i - M_0)(x_i - M_0)^T \right] w$$

$$\sigma_1^2 = w^T \left[\sum_{i=1}^{n_1} (x_i - M_1)(x_i - M_1)^T \right] w$$

$$S_w = \sum_{i=1}^{n_0} (x_i - M_0)(x_i - M_0)^T + \sum_{i=1}^{n_1} (x_i - M_1)(x_i - M_1)^T$$

$$J(w) = \frac{w^T S_b w}{w^T S_w w}$$

$$\frac{\partial J}{\partial w} = \frac{2 S_b w}{(w^T S_w w)} - \frac{(w^T S_b w) 2 w S_w}{(w^T S_w w)^2} = 0$$

Solve

$$\Rightarrow S_w w = \lambda S_b w$$

$$\Rightarrow W = \lambda S_n^{-1} S_B W$$